

Huan Wang

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 🎓 Google Scholar

Research Interests

Machine Learning, Natural Language Processing, Computer Vision, Large Action Models, AI Agents, Reasoning, and Planning.

Education

- Ph.D. Yale University**, Computer Science 2008–2013
- Advisor: [Prof. Daniel Spielman](#), [Prof. John Wright](#)
 - Research on clustering, regression, and dictionary learning
- M.Phil. The Chinese University of Hong Kong**, Information Engineering 2005–2007
- Advisor: [Prof. Xiaoou Tang](#), [Prof. Shuicheng Yan](#), [Prof. Jianzhuang Liu](#)
 - Research on face recognition, manifold learning, subspace learning, semi-supervised learning
- B.Eng. Zhejiang University**, Information Engineering, Chu Kochen Honors College 2000–2004

Professional Experience

- Senior Director**, Salesforce Research Palo Alto, CA
Feb 2025 – Present
- Large Action Models (xLAM), AI Agents, Reasoning, Planning, SWE Agents, Agent-force
- Director**, Salesforce Research Palo Alto, CA
Nov 2021 – Feb 2025
- Led teams in AI for Operational Research (AIOps), AI for Software, Conversational AI, Time-Series Anomaly Detection, Uncertainty Estimation, and Data Hardness Evaluation
- Senior Manager**, Salesforce Research Palo Alto, CA
Nov 2019 – Nov 2021
- Deep learning theory and applications, reinforcement learning, multi-task learning, language modeling, multilingual NER, knowledge graph
- Senior Research Scientist**, Salesforce Research Palo Alto, CA
2018 – 2019
- Deep learning research in large-scale language modeling and vision-language integration
- Senior Applied Researcher**, Microsoft AI Research Sunnyvale, CA
2015 – 2018
- Developed deep learning systems for recommendation, ranking, and intelligent Q&A
 - Improved web and local search relevance using neural embeddings
- Research Scientist**, Yahoo! Labs New York, NY
2013 – 2015
- Designed large-scale ML algorithms for search ads prediction and account security prediction
 - Leveraged Hadoop, Spark, and Storm for data processing

Adjunct Professor , New York University (<i>concurrent</i>)	New York, NY
• Taught the "Machine Learning" class.	2012 – 2013
Adjunct Professor , Baruch College (<i>concurrent</i>)	New York, NY
• Taught the "Algorithm Design" class.	2012 – 2013
Research Intern , Microsoft Research	Redmond, WA
• Projects on anomaly detection and time series modeling	2011 – 2011
Research Intern , Microsoft Research Asia	Beijing, China
• Projects on anomaly detection, image modeling, and dictionary learning	2010 – 2010

Honors and Awards

- Best Paper Award, Conference on Learning Theory (COLT), 2012
- Award of Excellence – Stars of Tomorrow, Microsoft Research, Asia
- Bachelor's Degree with Honors, Zhejiang University

Representative Publications

Full publication list: [Google Scholar](#)

Large Language Models (LLMs)

- [APIGen-MT: Agentic Pipeline for Multi-Turn Data Generation via Simulated Agent-Human Interplay.](#) [NeurIPS Datasets & Benchmarks Track, 2025.](#) (*co-corresponding author*). [\[Data\]](#), [\[Model\]](#)
- [APIGen: Automated Pipeline for Generating Verifiable and Diverse Function-Calling Datasets.](#) [NeurIPS, 2024.](#) [\[Data\]](#), [\[Model\]](#)
- [xLAM: A Family of Large Action Models to Empower AI Agent Systems.](#) [NAACL, 2024.](#) (*co-corresponding author*). [\[Code\]](#)
- [Retroformer: Retrospective Large Language Agents with Policy Gradient Optimization.](#) [ICLR, 2024.](#) (*co-corresponding author*).
- [CodeGen: An Open Large Language Model for Code with Multi-Turn Program Synthesis.](#) [ICLR, 2023.](#) [\[Code\]](#)

AI Agents and Multi-Agent Systems

- [BOLAA: Benchmarking and Orchestrating LLM-Augmented Autonomous Agents.](#) [arXiv, 2023.](#) [\[Code\]](#)
- [AgentLite: A Lightweight Library for Building and Advancing Task-Oriented LLM Agent System.](#) [arXiv, 2024.](#) [\[Code\]](#)
- [CRMArena: Understanding the Capacity of LLM Agents to Perform Professional CRM Tasks in Realistic Environments.](#) [NAACL, 2025.](#) [\[Code\]](#)
- [ActionStudio: A Lightweight Framework for Data and Training of Large Action Models.](#) [EMNLP, 2025.](#)
- [UserRL: Training Interactive User-Centric Agent via Reinforcement Learning.](#) [arXiv, 2025.](#) [\[Code\]](#)
- [Enterprise Deep Research: Steerable Multi-Agent Deep Research for Enterprise Analytics.](#) [arXiv, 2025.](#)
- [MCP Eval: Automatic MCP-based Deep Evaluation for AI Agent Models.](#) [arXiv, 2025.](#) [\[Code\]](#)
- [REX: Rapid Exploration and eXploitation for AI Agents.](#) [arXiv, 2023.](#)

LLM Reasoning, Routing, and Software Engineering

- [Language Models are Hidden Reasoners: Unlocking Latent Reasoning Capabilities via Self-Rewarding.](#) [arXiv, 2024.](#)
- [PRACT: Optimizing Principled Reasoning and Acting of LLM Agent.](#) [CoNLL, 2024.](#)
- [LATTE: Learning to Think with Vision Specialists.](#) [arXiv, 2024.](#)
- [ToolLibGen: Scalable Automatic Tool Creation and Aggregation for LLM Reasoning.](#) [arXiv, 2025.](#)
- [xRouter: Training Cost-Aware LLMs Orchestration System via Reinforcement Learning.](#) [arXiv, 2025.](#)

- [LoCoBench: A Benchmark for Long-Context Large Language Models in Complex Software Engineering](#). [arXiv](#), 2025. [\[Code\]](#) [🔗](#)

Theory of Deep Learning and In-Context Learning

- [Transformers as Statisticians: Provable In-Context Learning with In-Context Algorithm Selection](#). [NeurIPS](#), 2023.
- [How do Transformers Learn In-Context Beyond Simple Functions? A Case Study on Learning with Representations](#). [ICLR](#), 2024. [🔗](#)
- [How Important is the Train-Validation Split in Meta-Learning?](#) [ICML](#), 2021. [🔗](#)
- [Catastrophic Fisher Explosion: Early Phase Fisher Matrix Impacts Generalization](#). [ICML](#), 2021. [🔗](#)
- [Ensemble of Averages: Improving Model Selection and Boosting Performance in Domain Generalization](#). [NeurIPS](#), 2022. [🔗](#)

Reinforcement Learning

- [Policy Finetuning: Bridging Sample-Efficient Offline and Online Reinforcement Learning](#). [NeurIPS](#), 2021. [🔗](#)
- [Sample-Efficient Learning of Stackelberg Equilibria in General-Sum Games](#). [NeurIPS](#), 2021. [🔗](#)
- [WarpDrive: Extremely Fast End-to-End Deep Multi-Agent Reinforcement Learning on a GPU](#). [arXiv](#), 2021. [🔗](#)
- [On the Generalization Gap in Reparameterizable Reinforcement Learning](#). [ICML](#), 2019. [🔗](#)

Uncertainty Estimation & Reliability

- [Improved Online Conformal Prediction via Strongly Adaptive Online Learning](#). [ICML](#), 2023. [🔗](#)
- [Understanding the Under-Coverage Bias in Uncertainty Estimation](#). [NeurIPS](#), 2021. [🔗](#)
- [Don't Just Blame Over-parametrization for Over-confidence: Theoretical Analysis of Calibration in Binary Classification](#). [ICML](#), 2021. [🔗](#)

Time Series, Causality and Neural Editing

- [Merlion: End-to-End Machine Learning for Time Series](#). [JMLR](#), 2023. [🔗](#)
- [Causal Layering via Conditional Entropy](#). [PMLR](#), 2024. [🔗](#)
- [Salesforce CausalAI Library: A Fast and Scalable Framework for Causal Analysis of Time Series and Tabular Data](#). [arXiv](#), 2023. [🔗](#)
- [On the Unlikelihood of D-Separation](#). [arXiv](#), 2023. [🔗](#)
- [Editing Arbitrary Propositions in LLMs without Subject Label](#). [arXiv](#), 2024. [🔗](#)

Earlier Work in Machine Learning and Sparse Representation

- [Exact Recovery of Sparsely-Used Dictionaries](#). [COLT](#), 2012. **Best Paper Award**. [🔗](#)
- [Adaptive Dropout with Rademacher Complexity Regularization](#). [ICLR](#), 2018. [🔗](#)
- [Trace ratio vs. ratio trace for dimensionality reduction](#). [CVPR](#), 2007. [🔗](#)

Open Source Projects

Large Action Models and AI Agents

- [xLAM & ActionStudio](#) [🔗](#): A family of large action models for AI agent systems, plus a lightweight framework for action-model data and training.
- [AgentLite](#) [🔗](#): A lightweight library for building task-oriented LLM agent systems.
- [BOLAA](#) [🔗](#): Benchmarking and orchestrating LLM-augmented autonomous agents.
- [CRMArena](#) [🔗](#): A benchmark for evaluating LLM agents on professional CRM tasks.
- [MCPEval](#) [🔗](#): Automatic MCP-based deep evaluation for AI agent models.
- [Enterprise Deep Research](#) [🔗](#): Steerable multi-agent deep research for enterprise analytics.
- [PersonaBench](#) [🔗](#): A benchmark for persona-based conversation systems.
- [MobileAIBench](#) [🔗](#): A benchmark for evaluating AI systems on mobile devices.

Function Calling and API Generation

- [APIGen](#) [🔗](#): Automated pipeline for generating verifiable and diverse function-calling datasets.

- [APIGen-MT](#): Agentic pipeline for multi-turn data generation via simulated agent-human interplay.

Code Generation and Software Engineering

- [CodeGen](#): A family of open-source models for code generation.
- [CoDA](#): Coding language models via diffusion adaptation.
- [LoCoBench](#): A benchmark for long-context large language models in complex software engineering.
- [Diversity Empowers Intelligence](#): Integrating expertise of software engineering agents.

Vision and Multimodal Models

- [LATTE](#): Learning to think with vision specialists.
- [xGen-MM \(BLIP3\)](#): Multimodal generative models. [\[Code\]](#)
- [UniControl](#): A unified diffusion model for controllable image generation.
- [Hive](#): Harnessing Human Feedback for Instructional Visual Editing.

Reinforcement Learning and Optimization

- [WarpDrive](#): Extremely fast end-to-end deep multi-agent reinforcement learning on GPU.
- [UserRL](#): User-centric reinforcement learning for interactive LLM agents.
- [UserBench](#): An interactive gym environment for user-centric agents.
- [xRouter](#): Training cost-aware LLM orchestration systems via reinforcement learning.

Prompt Optimization

- [Promptomatix](#): A powerful framework for LLM prompt optimization.
- [Retroformer](#): Retrospective large language agents with policy gradient optimization.

Time Series and Causal Learning

- [Merlion](#): An easy-to-use library for time series anomaly detection and forecasting. [\[Blog\]](#)
- [CausalAI](#): A library for causal inference and causal discovery.

Dialog and Conversation Systems

- [Converse](#): A framework for conversational AI applications.
- [DialogStudio](#): A comprehensive dialogue understanding benchmark and toolkit.

Industry Impact

- Research contributions to Salesforce **Agentforce** platform: xLAM models powering AI agent systems, APIGen-driven function-calling capabilities, and reasoning frameworks for enterprise AI agents.
- Open-sourced models and datasets adopted broadly in the community: xLAM family of action models, APIGen-MT-5k, xLAM-function-calling-60k (millions of downloads on Hugging Face).

Professional Service

- Area Chair / Senior Reviewer: NeurIPS, ICML, ICLR (recent years)
- Reviewer: NeurIPS, ICML, ICLR, AAAI, ACL, EMNLP, NAACL, JMLR, TPAMI

Invited Talks

- Invited talks on Large Action Models, AI Agents, and APIGen at academic and industry venues.

Skills

Technical: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Algorithms, AI Agents, Data Mining

Programming: Python

Frameworks & Tools: PyTorch, TensorFlow, Spark, Hadoop

Languages: English, Chinese